

P8109 Statistical Inference – Homework 2

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Problem 1

Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$ random variables. Find the MLE of λ .

The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{X_i}}{X_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n X_i}}{\prod_{i=1}^n X_i!}.$$

The log-likelihood is

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^n X_i \right) \log \lambda - \sum_{i=1}^n \log(X_i!).$$

Taking the derivative and setting it to zero:

$$\frac{d\ell}{d\lambda} = -n + \frac{\sum_{i=1}^n X_i}{\lambda} = 0.$$

Solving gives $\hat{\lambda}_{\text{MLE}} = \bar{X}$.

The second derivative $d^2\ell/d\lambda^2 = -\sum X_i/\lambda^2 < 0$ confirms this is a maximum.

Problem 2

Consider a sequence of i.i.d. random variables from a $N(\mu, \sigma^2)$ distribution. Let $m = \frac{1}{n} \sum_{i=1}^n X_i^2$.

(a) Given that $\sigma^2 = \mu$, show that the MLE of μ is $\hat{\mu} = \frac{-1 + \sqrt{1+4m}}{2}$.

Proof. When $\sigma^2 = \mu$, the log-likelihood is

$$\ell(\mu) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \mu - \frac{1}{2\mu} \sum_{i=1}^n (X_i - \mu)^2.$$

Expanding $(X_i - \mu)^2 = X_i^2 - 2\mu X_i + \mu^2$:

$$\ell(\mu) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \mu - \frac{nm}{2\mu} + n\bar{X} - \frac{n\mu}{2}.$$

Taking the derivative:

$$\frac{d\ell}{d\mu} = -\frac{n}{2\mu} + \frac{nm}{2\mu^2} - \frac{n}{2} = 0.$$

Multiplying by $2\mu^2/n$:

$$-\mu + m - \mu^2 = 0 \quad \Rightarrow \quad \mu^2 + \mu - m = 0.$$

By the quadratic formula: $\mu = \frac{-1 \pm \sqrt{1+4m}}{2}$. Since $\mu > 0$ (as variance must be positive), we take the positive root:

$$\hat{\mu} = \frac{-1 + \sqrt{1+4m}}{2}.$$

(b) Given that $\sigma = \mu$, show that the MLE of μ is $\hat{\mu} = \frac{-\bar{X} + \sqrt{\bar{X}^2 + 4m}}{2}$.

Proof. When $\sigma = \mu$, we have $\sigma^2 = \mu^2$. The log-likelihood is

$$\ell(\mu) = -\frac{n}{2} \log(2\pi) - n \log \mu - \frac{1}{2\mu^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Expanding:

$$\ell(\mu) = -\frac{n}{2} \log(2\pi) - n \log \mu - \frac{nm}{2\mu^2} + \frac{n\bar{X}}{\mu} - \frac{n}{2}.$$

Taking the derivative:

$$\frac{d\ell}{d\mu} = -\frac{n}{\mu} + \frac{nm}{\mu^3} - \frac{n\bar{X}}{\mu^2} = 0.$$

Multiplying by μ^3/n :

$$-\mu^2 + m - \bar{X}\mu = 0 \quad \Rightarrow \quad \mu^2 + \bar{X}\mu - m = 0.$$

By the quadratic formula: $\mu = \frac{-\bar{X} \pm \sqrt{\bar{X}^2 + 4m}}{2}$. Taking the positive root:

$$\hat{\mu} = \frac{-\bar{X} + \sqrt{\bar{X}^2 + 4m}}{2}.$$

Problem 3

Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, 1)$ distribution.

(a) Show that $T = \frac{1}{n} \sum_{i=1}^n X_i^2 - 1$ is an unbiased estimator of μ^2 .

Proof. For $X_i \sim N(\mu, 1)$, we have $E[X_i^2] = \text{Var}(X_i) + (E[X_i])^2 = 1 + \mu^2$. Therefore,

$$E[T] = E\left[\frac{1}{n} \sum_{i=1}^n X_i^2 - 1\right] = \frac{1}{n} \sum_{i=1}^n E[X_i^2] - 1 = (1 + \mu^2) - 1 = \mu^2.$$

Hence T is unbiased for μ^2 .

(b) Explain why $2T$ is not an appropriate estimator of $2\mu^2$.

Although $E[2T] = 2\mu^2$, the estimator $2T$ is not appropriate because T (and hence $2T$) can take negative values. Since $2\mu^2 \geq 0$ for all μ , an estimator that can be negative violates the natural constraint of the parameter space. For instance, if $\mu = 0$ and $n = 1$, then $T = X_1^2 - 1$ has a positive probability of being negative when $|X_1| < 1$.

(c) Give a simple proof that T is a biased estimator of μ .

Proof. From part (a), $E[T] = \mu^2$. For T to be unbiased for μ , we would need $E[T] = \mu$, i.e., $\mu^2 = \mu$ for all μ . This only holds when $\mu \in \{0, 1\}$. For any other value of μ , we have $E[T] = \mu^2 \neq \mu$, so T is biased for μ .

Problem 4

(a) Define what it means for T_n to be a consistent estimator of a parameter θ .

Definition. T_n is a consistent estimator of θ if $T_n \xrightarrow{p} \theta$ as $n \rightarrow \infty$, i.e., for any $\varepsilon > 0$,

$$P(|T_n - \theta| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

(b) Given T_n is unbiased for θ , use Chebyshev's inequality to show that T_n is consistent for θ if $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$.

Proof. By Chebyshev's inequality, for any $\varepsilon > 0$:

$$P(|T_n - E[T_n]| > \varepsilon) \leq \frac{\text{Var}(T_n)}{\varepsilon^2}.$$

Since T_n is unbiased, $E[T_n] = \theta$. Thus:

$$P(|T_n - \theta| > \varepsilon) \leq \frac{\text{Var}(T_n)}{\varepsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

provided $\text{Var}(T_n) \rightarrow 0$. Therefore $T_n \xrightarrow{p} \theta$, i.e., T_n is consistent for θ .

Problem 5

Let $X_1, \dots, X_n$ be i.i.d. $N(\theta, 1)$ random variables.

(a) Obtain the CRLB for any unbiased estimator of θ .

The log-likelihood for one observation is

$$\log f(x; \theta) = -\frac{1}{2} \log(2\pi) - \frac{(x - \theta)^2}{2}.$$

The score function is $S_1(\theta) = \frac{\partial}{\partial \theta} \log f = x - \theta$.

The Fisher information for one observation is

$$I_1(\theta) = E[S_1(\theta)^2] = E[(X - \theta)^2] = \text{Var}(X) = 1,$$

where the last equality holds because $E[X] = \theta$. For n i.i.d. observations: $I_n(\theta) = nI_1(\theta) = n$.

The CRLB for any unbiased estimator T of θ is:

$$\text{Var}(T) \geq \frac{1}{I_n(\theta)} = \frac{1}{n}.$$

(b) By writing the score function in suitable form, identify the function $\tau(\theta)$ for which an MVBU estimator exists. Write down the latter estimator.

The total score function is

$$S(\theta) = \sum_{i=1}^n (X_i - \theta) = n\bar{X} - n\theta.$$

This can be written in the form $S(\theta|\mathbf{X}) = g_1(\theta)T(\mathbf{X}) + g_2(\theta)$ as:

$$S(\theta) = n \cdot \bar{X} + (-n\theta),$$

where $g_1(\theta) = n$, $T(\mathbf{X}) = \bar{X}$, and $g_2(\theta) = -n\theta$.

By the Cram'er-Rao theorem (Theorem 4.4), since the score function is a linear combination of $\bar{X}$, the MVBU estimator exists. From the theorem, the MVBU estimator is $T(\mathbf{X}) = \bar{X}$, and the corresponding parameter function is

$$\tau(\theta) = E[\bar{X}] = \theta.$$

Therefore, $\tau(\theta) = \theta$ and the MVBU estimator is $\bar{X}$.

(c) Write down the UMVUE for θ .

Since $\bar{X}$ achieves the CRLB, it is the MVBU (minimum variance bound unbiased) estimator and hence is the UMVUE for θ :

$$\text{UMVUE} = \bar{X}.$$

(d) Obtain the CRLB for any unbiased estimator of 2θ .

Let $\tau(\theta) = 2\theta$. Then $\tau'(\theta) = 2$. By the Cram'er-Rao inequality:

$$\text{Var}(T) \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)} = \frac{4}{n}.$$

$$\text{CRLB for } 2\theta = \frac{4}{n}.$$

(e) By writing the score function in suitable form, identify the MVBU estimator of 2θ if it exists.

The score function is $S(\theta) = n\bar{X} - n\theta$. We can rewrite this in the form $S(\theta|\mathbf{X}) = g_1(\theta)T(\mathbf{X}) + g_2(\theta)$ as:

$$S(\theta) = \frac{n}{2} \cdot (2\bar{X}) + (-n\theta),$$

where $g_1(\theta) = n/2$, $T(\mathbf{X}) = 2\bar{X}$, and $g_2(\theta) = -n\theta$.

By the Cram'er-Rao theorem (Theorem 4.4), since the score function is a linear combination of $2\bar{X}$, the MVBU estimator exists. The MVBU estimator is $T(\mathbf{X}) = 2\bar{X}$, and the corresponding parameter function is

$$\tau(\theta) = E[2\bar{X}] = 2\theta.$$

Therefore, the MVBU estimator of 2θ is $2\bar{X}$.

We can verify: $E[2\bar{X}] = 2\theta$ (unbiased), and $\text{Var}(2\bar{X}) = 4 \text{Var}(\bar{X}) = 4/n$, which equals the CRLB.

Problem 6

Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$ random variables. We wish to estimate $p = P(X_j = 0)$.

(a) Find an expression for p in terms of λ .

For $X \sim \text{Poisson}(\lambda)$:

$$p = P(X = 0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda}.$$

(b) Find the CRLB in terms of p for any unbiased estimator of p .

For $X \sim \text{Poisson}(\lambda)$, the log-likelihood is $\log f(x; \lambda) = -\lambda + x \log \lambda - \log(x!)$. The score is $\frac{\partial}{\partial \lambda} \log f = \frac{x}{\lambda} - 1$, and

$$I_1(\lambda) = -E \left[\frac{\partial^2}{\partial \lambda^2} \log f \right] = -E \left[-\frac{X}{\lambda^2} \right] = \frac{E[X]}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

Thus $I_n(\lambda) = n/\lambda$.

Since $p = e^{-\lambda}$, we have $\frac{dp}{d\lambda} = -e^{-\lambda} = -p$.

By the Cram'er-Rao inequality:

$$\text{Var}(T) \geq \frac{(dp/d\lambda)^2}{I_n(\lambda)} = \frac{p^2}{n/\lambda} = \frac{p^2 \lambda}{n}.$$

Since $\lambda = -\log p$, the CRLB in terms of p is:

$$\text{CRLB} = \frac{-p^2 \log p}{n}.$$

(c) Let $Y_j = \begin{cases} 1, & \text{if } X_j = 0, \\ 0, & \text{otherwise.} \end{cases}$ Find EY_j and use this result to write down an unbiased estimator T of p .

We have

$$E[Y_j] = P(X_j = 0) = e^{-\lambda} = p.$$

Therefore, an unbiased estimator of p is

$$T = \bar{Y} = \frac{1}{n} \sum_{j=1}^n Y_j.$$

(d) Find $\text{Var}(T)$ (in terms of p).

Since $Y_j \sim \text{Bernoulli}(p)$, we have $\text{Var}(Y_j) = p(1-p)$. Therefore,

$$\text{Var}(T) = \text{Var}(\bar{Y}) = \frac{\text{Var}(Y_1)}{n} = \frac{p(1-p)}{n}.$$

(e) By graphing suitable functions or otherwise, determine if T is the MVBU estimator.

T is MVBU if and only if $\text{Var}(T) = \text{CRLB}$, i.e.,

$$\frac{p(1-p)}{n} = \frac{-p^2 \log p}{n}.$$

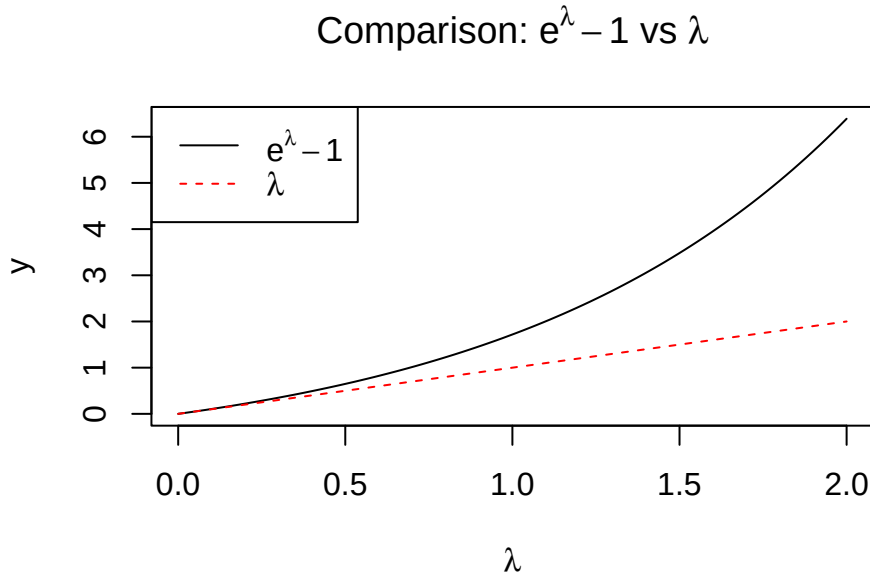
This simplifies to:

$$\frac{1}{p} - 1 = -\log p.$$

Let $\lambda = -\log p$ (so $p = e^{-\lambda}$). The condition becomes:

$$e^\lambda - 1 = \lambda.$$

```
lambda_vals <- seq(0, 2, length.out = 100)
plot(lambda_vals, exp(lambda_vals) - 1, type = "l",
     ylab = "y", xlab = expression(lambda),
     main = expression(paste("Comparison: ", e^lambda - 1, " vs ", lambda)))
lines(lambda_vals, lambda_vals, lty = 2, col = "red")
legend("topleft", legend = c(expression(e^lambda - 1), expression(lambda)),
     lty = c(1, 2), col = c("black", "red"))
```



The curves intersect only at $\lambda = 0$ (i.e., $p = 1$), which is a degenerate case. For all $\lambda > 0$, we have $e^\lambda - 1 > \lambda$ by the strict convexity of e^λ .

This means $\text{Var}(T) > \text{CRLB}$ for all $p \in (0, 1)$. Therefore, T does **not** achieve the CRLB and is **not** the MVBU estimator.